

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2021-P2-Model)** **Date: 08-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-15** **Max. Marks: 180**

08-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2021-P2)_GTA-15_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCE-2021-P2-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 7 – 12)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 13 – 16)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 25)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 26 – 31)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 32 – 35)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 44)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 45 – 50)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 51 – 54)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

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PHYSICS

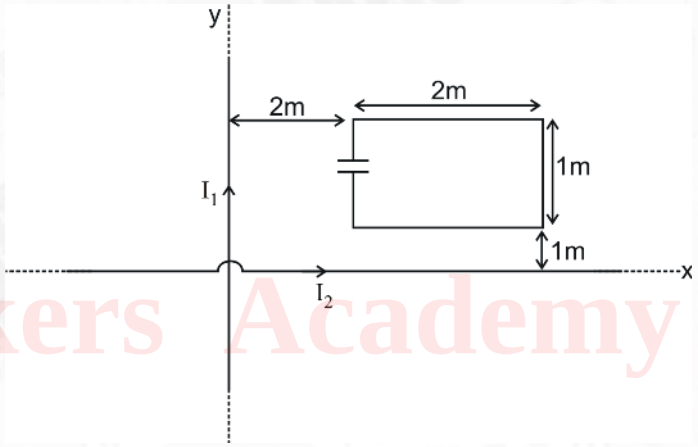
Max. Marks: 60

SECTION-1(Maximum Marks: 24)

One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : −2 In all other cases.

1. Two infinite current carrying wires lie overlapping each other along x & y axes carrying current $I_2 = I_0 \cos\left(\omega t + \frac{\pi}{4}\right)$ and $I_1 = 2I_0 \sin\left(\omega t + \frac{\pi}{4}\right)$. A rectangular frame of negligible resistance with capacitor of capacitance C is placed in the xy plane as shown. Select correct options



A) Maximum emf induced in frame is $\frac{\mu_0 I_0 \omega \ell n 2}{2\pi}$

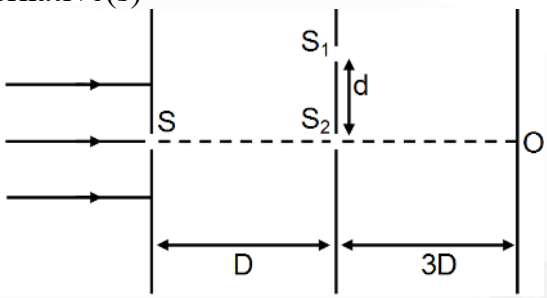
B) Maximum emf induced in frame is $\frac{\sqrt{2}\mu_0 I_0 \omega \ell n 2}{\pi}$

C) Maximum charge on capacitor is $\frac{\sqrt{2}\mu_0 I_0 \omega C \ell n 2}{\pi}$

D) Current in frame at $t = \frac{\pi}{\omega}$ is zero

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2. In the YDSE setup shown in figure a monochromatic light of wavelength λ is incident on slit S. Distance D is very large as compared to separation d between slits S_1 and S_2. Choose the correct alternative(s)



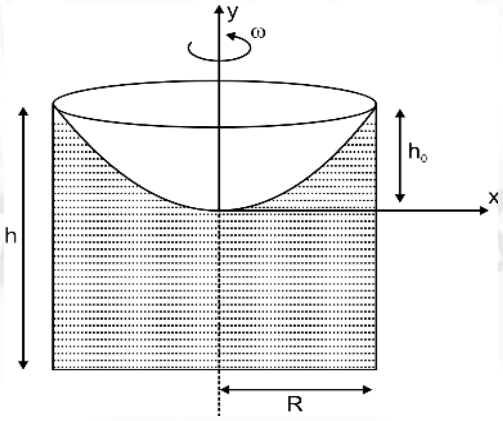
A) Minimum value of d for which there is a bright fringe at O is $\sqrt{\frac{2\lambda D}{3}}$

B) Minimum value of d for which there is a bright fringe at O is $\sqrt{\frac{3\lambda D}{2}}$

C) Position of central bright fringe for this minimum value of d is $\sqrt{\frac{2\lambda D}{3}}$ above O

D) Position of central bright fringe for this minimum value of d is $\sqrt{6\lambda D}$ above O

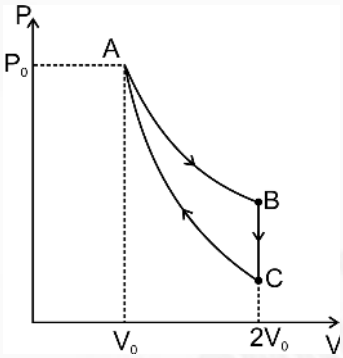
3. An ideal liquid of density ρ is filled in a cylindrical container upto $(3/4)^{th}$ height. Now the liquid is being rotated about vertical axis passing through its axis of symmetry with constant angular velocity ω, such that liquid is on the verge of falling out of the container and free surface of liquid forms a paraboloid. Whole situation is shown in the figure. Choose the correct option(s). [P = pressure]



A) $h_0 = \frac{h}{4}$ **B)** $\omega = \sqrt{\frac{gh}{R^2}}$ **C)** $\left|\frac{dP}{dx}\right| = \rho\omega^2 x$ **D)** $\left|\frac{dP}{dy}\right| = \rho g$

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4. 1 mol He gas undergoes a cyclic process ABCA as shown in the figure. Process AB, BC & CA are isothermal, isochoric and adiabatic respectively. Temperature of gas at point A is T_0. On the basis of given information identify correct statements. $(2^{2/3} = 1.5)$ and $(\ln 2 = 0.7)$



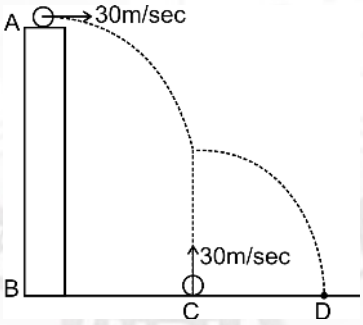
A) Change in internal energy in process BC is - $RT_0 / 2$

B) Work done by gas in process CA is $-RT_0 / 2$

C) Work done by gas in cyclic process is $RT_0 / 5$

D) Efficiency of cyclic process is 2/7.

5. A particle of mass m is thrown horizontally from highest point A of tower AB with speed 30m/sec. At the same time another particle of mass 2m is thrown vertically upward from point C on ground with same speed 30 m/sec. Both particles strike in air and stuck to each other after two seconds of projection. Combined mass fall on ground at point D. On the basis of this information which of the option(s) given below is/are correct? (take g = $10 m/sec^2$ and $\sqrt{2} = 1.4$)



A) Height of tower AB and distance BC, both are 60 meter.

B) After collision combined mass move with speed 20m/sec.

C) Distance BD is equal to 88 meter.

D) Time of flight of particle A is $2\sqrt{3}$ sec (If there is no collision)

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6. A particle of specific charge α is projected from origin at t = 0 with velocity $\vec{v} = v_0\hat{i} + v_0\hat{k}$ in magnetic field $\vec{B} = -B_0\hat{k}$. Mass of the particle is 1 SI unit. On the basis of given information find correct options.

A) Velocity of particle at $t = \frac{\pi}{\alpha B_0}$ is $-v_0\left(\hat{i} - \hat{k}\right)$.

B) Speed of the particle at $t = \frac{\pi}{\alpha B_0}$ is $\frac{\sqrt{5}}{2}v_0$.

C) Distance travelled by particle upto time $t = \frac{2\pi}{\alpha B_0}$ is $\frac{2\sqrt{2}v_0\pi}{\alpha B_0}$

D) Displacement of particle upto time $t = \frac{3\pi}{\alpha B_0}$ is $\frac{3\pi v_0}{\alpha B_0}$

SECTION-2(Maximum Marks: 12)

Paragraph with Numerical

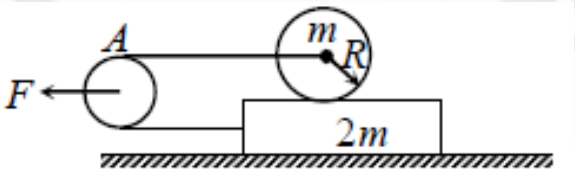
- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 7 and 8

Question Stem

A solid cylinder of mass m and radius R is kept at rest on a plank of mass $2m$ lying on a smooth horizontal surface. Massless and inextensible string connecting cylinder to the plank is passing over a massless pulley. The friction between the cylinder and the plank is sufficient to prevent slipping. Pulley A is pulled with a constant horizontal force F.
[F =100N, m = 1kg]



7. Acceleration of cylinder with respect to earth is (in $m/\sec^2$)

8. Acceleration of plank with respect to earth is (in $m/\sec^2$)

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Question Stem for Question Nos. 9 and 10

Question Stem

A light of wavelength λ is incident on a metal sheet of work function $\phi = 2eV$. The wavelength λ varies with time as $\lambda = 3000 + 40t$, where λ is in $\overset{o}{A}$ and t is in second. The power incident on metal sheet I constant at 100 W. This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on, the λ start from initial value of 3000 $\overset{o}{A}$. The metal plate is grounded and electron clouding is negligible. The efficiency of photoemission is 1% (hc = 12400 eV $\overset{o}{A}$).

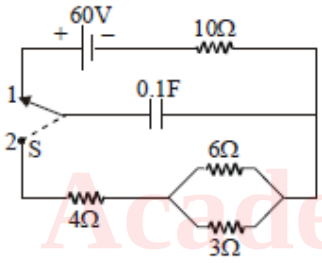
9. The time after which photoemission will stop is (in sec)

10. If the total number of photoelectrons ejected in one hour is $N \times 10^{20}$, value of 'N' is

Question Stem for Question Nos. 11 and 12

Question Stem

A two way switch S is used in the circuit shown in figure. First, the capacitor is charged by putting the switch in position 1. After a long time, the switch is put in position 2.



11. What is the heat generated across 4Ω resistor (in joule)?

12. What is the heat generated across 10Ω resistor in the entire process (in joule)?

SECTION-3(Maximum Marks: 12)

Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-I

If a plane simple harmonic wave is traveling towards positive x axis, its wave function is given by

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$S = S_0 \sin(\omega t - kx)$

Here S is longitudinal displacement of any particle. If we take a plane parallel to yz plane all particles in this plane have same phase. This plane is called wavefront. If a plane wave is traveling along direction $\hat{n}$, its wave function can be written as :

$S = S_0 \sin(\omega t - k\hat{n} \cdot \vec{r})$

Let's call $k\hat{n} = \vec{k}$ (a wave vector)

$S = S_0 \sin\left(\omega t - \vec{k} \cdot \vec{r}\right)$

Wave vector points in direction of phase velocity. In other words, the wave vector points in the normal direction to the wavefront.

13. Consider a wave function:

$S = S_0 \sin(\omega t - 2x + 2y + z)$ Here all variables are in SI units.

Select correct Statement:

A) Wavelength of the wave is $\frac{2\pi}{3}$

B) Particle (0, 0, 0) and (1, 1, 2) are on same wavefront.

C) The wave vector is at angle $\cos^{-1}\left(\frac{1}{3}\right)$ from x-axis.

D) The wave vector is at equal angle from positive x-axis and negative z-axis.

14. Consider following time wave function.

$S_1 = S_0 \sin(\omega t - 2x + 2y + z)$

$S_2 = S_0 \sin(\omega t + 2x - 2y - z)$

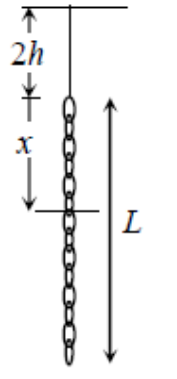
An observer is traveling towards +x axis with speed π m/s. The beat frequency observed by the observer is :

A) $\frac{1}{2}$ **B) 2** **C) 1** **D)** $\frac{3}{2}$

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Paragraph-II

A uniform chain of length L is suspended in equilibrium by an elastic string of natural length h as shown. The chain is then suddenly cut at point A which is x distance below the upper end of chain.



15. If x = L/3, the velocity of suspended part of chain when it rises by $\frac{h}{2}$ from its equilibrium position is

A) $\sqrt{\frac{gh}{3}}$ **B)** $\sqrt{\frac{4}{3}gh}$ **C)** $\sqrt{\frac{5}{4}gh}$ **D)** O

16 To allow the elastic string to become slack, the minimum value of x is

A) $\frac{L}{2}$ **B)** $\frac{L}{3}$ **C)** $\frac{L}{4}$ **D)** $\frac{2L}{3}$

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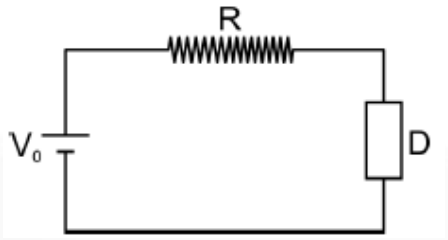
SECTION-4(Maximum Marks: 12)
Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

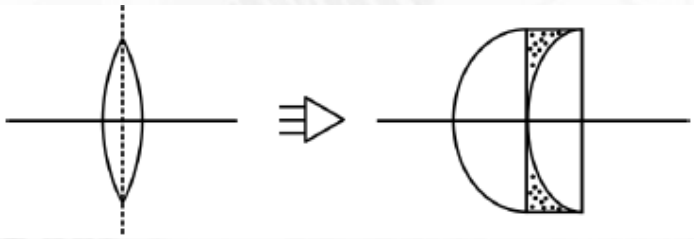
Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

17. A resistance R and nonlinear device D are connected with an ideal battery as shown in the given network. Current I through the non linear device varies with the voltage V according to the equation $I^2 = V \cdot K$ where K is positive constant. If current in the circuit is given by equation I = $\frac{-RK + \sqrt{R^2K^2 + xV_0K}}{2}$ then find x.

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18. An equiconvex lens of focal length 20cm is placed in air medium. It is divided into two parts about an axis ⊥ to principal axis. One part is inverted about axis of division of lens. Both parts are in contact with each other. Space between lenses is filled by a liquid of unknown refractive index. When light rays parallel to principal axis fall on this combination no deflection is produced in light rays by this combination. Find refractive index of liquid. {Refractive index of lens is $\frac{3}{2}$}



19. A particle is moving on x-axis and has potential energy $U = 5 - 20x + 2x^2$ joule where x is position of particle in meter. Particle is released at x = –2 meter. If mass of particle is 2 kg then maximum velocity of the particle is $k\sqrt{2}$ m/s. Find k.

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CHEMISTRY **Max. Marks: 60**

SECTION-1(Maximum Marks: 24)
One or More Type

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  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : −2 In all other cases.

20. Which of the following pairs do not contain equal number of atoms?

A) 11.2 mL of N_2 (at 1 atm and 0^oC) and 0.015 g of NO

B) 22.4 L of N_2O and 22.4 L of NO under identical conditions

C) 1 milli mole of HCl and 0.5 milli mole of H_2S

D) 1 mole of H_2O_2 and 1 mole of N_2O_4

21. $\left(\Delta_f^0 G\right)_{Cr_2O_3(s)} = -527 kJmol^{-1}$

$\left(\Delta_f^0 G\right)_{Al_2O_3(s)} = -827 kJmol^{-1}$

$\left(\Delta_f H^0\right)_{Cr_2O_3(s)} = -1100 kJ$

$\left(\Delta_f H^0\right)_{Al_2O_3(s)} = -1600 kJ$

Which is/are correct?

A) $Cr_2O_3(s)$ is reduced by Al(s) in given standard condition.

B) $Al_2O_3(s)$ is reduced by Cr(s) in given standard condition.

C) $\Delta_r H^o$ for the reduction of Cr_2O_3 by Al is -500 kJ

D) Reduction of Cr_2O_3 by Al is spontaneous at all temperature.

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22. Among the following solutions, which two solutions will show 'highest' or 'lowest' increase in temperature on mixing? Assume that initially all solutions at 25°C and density of all solutions are identical.

A) 0.4 M, 50 mL NaOH with 0.5 M, 100 mL HNO_3

B) 0.2 M , 75mL H_2SO_4 with 0.6 M, 25 mL NaOH

C) 0.3 M, 20 mL HBr with 0.3M, 80mL $Ba(OH)_2$

D) 0.9 M , 40 mL $Ba(OH)_2$ with 0.7 M , 60 mL HI

23. Three different solutions of oxidising agents $K_2Cr_2O_7, I_2$ and $KMnO_4$ is titrated separately with 0.19 g of $K_2S_2O_3$. The molarity of each oxidising agent is 0.1 M and the reactions are:

(P) $Cr_2O_7^{2-} + S_2O_3^{2-} \rightarrow Cr^{3+} + SO_4^{2-}$ (Q) $I_2 + S_2O_3^{2-} \rightarrow I^- + S_4O_6^{2-}$

(R) $MnO_4^- + S_2O_3^{2-} \rightarrow MnO_2 + SO_4^{2-}$

(Molecular weight of $K_2S_2O_3 = 190$, $K_2Cr_2O_7 = 294$, $KMnO_4 = 158$ and $I_2 = 254 g\, mol^{-1}$)

Which of the following statement(s) is (are) correct?

A) Volume of $KMnO_4$ is maximum.

B) Volume of I_2 used is minimum

C) Volume of $K_2Cr_2O_7$ used is maximum

D) n_f of $S_2O_3^{2-}$ in (P) and (R) are different because n_f of $Cr_2O_7^{2-}$ and MnO_4^- are different

24. Select correct for metallurgical process which is/are involve in extraction metal from galena.

A) Reduction of roasted ore with fresh galena

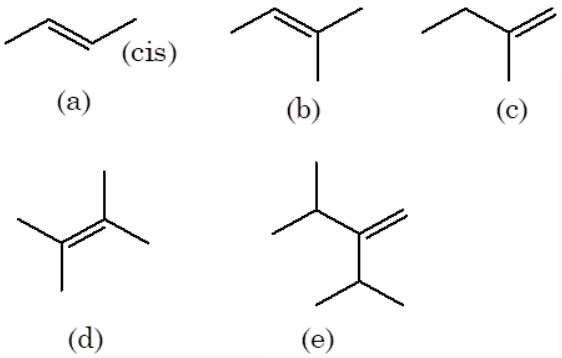
B) Copper impurities can be remove by liquation

C) Alkali metal cynide selectively prevent galena coming to the froth

D) Oxidation of silver impurity in molten metal by O_2

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25.



Select correct statements for all above alkenes.

A) Compound 'e' shows the highest heat of combustion.

B) Compound 'd' is the thermodynamically least reactive towards catalytic hydrogenation.

C) On ozonolysis using O_3 / Me_2S only 2 alkenes give smallest non–cyclic ketone.

D) Only 2 alkenes among these give chiral alcohol product(s) on reaction with B_2H_6/THF followed by H_2O_2 / OH^-

SECTION-2(Maximum Marks: 12)

Paragraph with Numerical

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
  Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 26 and 27

Question Stem

100 ml of 0.1M of H_3PO_4 is titrated with 0.05M NaOH solution till 2nd equivalence point. Then, resultant solution was mixed with 10ml of 0.5M HCl solution

$K_1 = 10^{-3}$ M

$K_2 = 10^{-8}$ M

$K_3 = 10^{-13}$ M

26. pH after HCl was added will be :

27. Solubility of compound $A(OH)_2$ in final solution will be $x \times 10^{-18}$ M then find the value of x

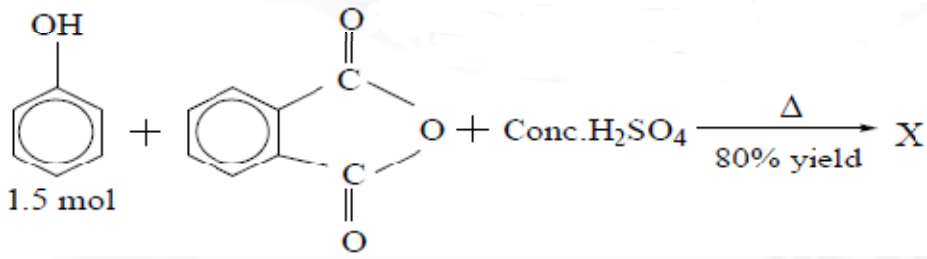
[Given : $K_{sp} A(OH)_2 = 4 \times 10^{-30} M^3$]

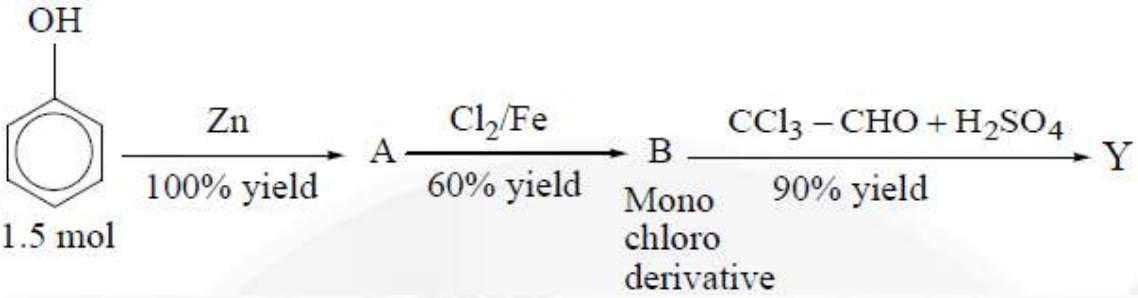
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Question Stem for Question Nos. 28 and 29

Question Stem

Phenol is one of the starting material of two commercially important compounds X and Y. they are prepared as follows. (atomic mass of Cl=35.5,O=16,S=32,H=1)





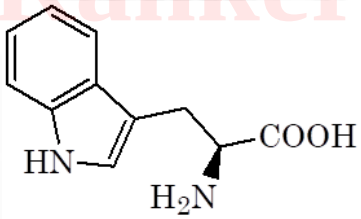
28. The mass in grams of 'X' obtained is _____

29. The mass in grams of 'Y' obtained is _____

Question Stem for Question Nos. 30 and 31

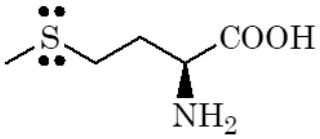
Question Stem

The following amino acids are used to form dipeptides :



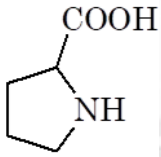
Tryptophan

(A)



Methionine

(B)



Proline

(C)

30. A dipeptide 'X' is formed which has maximum possible degree of unsaturation. The number of π –bonds in that dipeptide is 2n. Find the value of n.

31. Another dipeptide 'Y' with only one Sulphur atom and two nitrogen atoms is formed. Let 'P' is the number of oxygen atoms in 'Y' and 'Q' is the number of chiral centres in 'Y'. Then report the value of (P + Q)

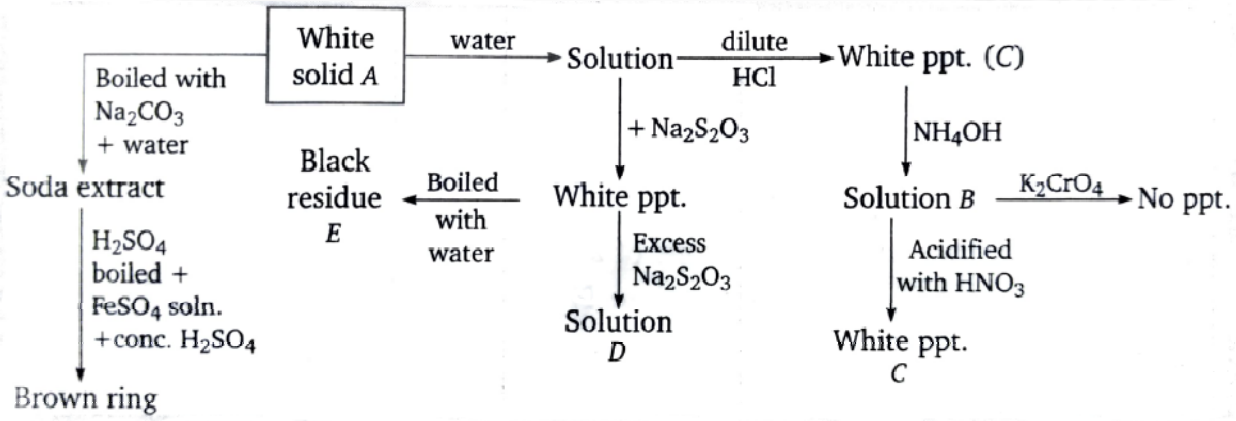
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SECTION-3(Maximum Marks: 12)
Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 *If ONLY the correct option is chosen;*
Zero Marks : 0 *If none of the options is chosen (i.e. the question is unanswered);*
Negative Marks : −1 *In all other cases.*

Paragraph-I:



32. The white solid A is a:

A) Chloride **B)** Nitrate **C)** Nitrite **D)** Bromide

33. The change from B to C involves the reaction:

A) $[Hg(NH_3)_4]^{2+} + 2H^+ + Cl^- \rightarrow Hg(NH_2)Cl\downarrow 3NH_4^+$

B) $[Pb(NH_3)_4]^{2+} + 4H^+ + 2Cl^- \rightarrow PbCl_2\downarrow + 4NH_4^+$

C) $[Pb(OH)_4]^{2-} + 4H^+ + 2Cl^- \rightarrow PbCl_2\downarrow + 4H_2O$

D) $\left[Ag(NH_3)_2\right]^+ + 2H^+ + Cl^{-1} \rightarrow AgCl\downarrow + 2NH_4^+$

Paragraph-II:

Some ligands not only donate their lone pair to the central metal atom but also accept the electron cloud from the central metal atom. This is known as synergic bonding.

34. In which of the following cases is the bond energy of C – O bond minimum?

A) Free CO molecule

B) Terminal CO group in a complex.

C) Double bridging CO in a complex.

D) Triply bridging CO^+ in a complex.

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35. Select the correct order for the stretching frequencies of C – O bond the following compounds.

(I) fac-Mo$(CO)_3(PF_3)_3$ (II) fac-Mo$(CO)_3(PCl_3)_3$

(III) fac-Mo$(CO)_3(PMe_3)_3$

A) I < II < III **B)** I > II > III **C)** I = II = III **D)** II > I > III

SECTION-4(Maximum Marks: 12)
Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

36. Given :

$$\lambda^0_{H^+} = 350 cm^2 mol^{-1}$$

$$\lambda^0_{A^-} = 50 cm^2 mol^{-1}$$

Molar conductivity of 0.038 M 'HA' solution is 20 Scm^2mol^{-1} at 298 K. What is the pH of 0.01 M NaA salt solution (Express your answer to nearest integer)

37. Find the number of combination(s) of reagents in which pink coloured solution or precipitate or product(s) is/are formed.

i) $MnCl_2 + H_2S(g) + HCl$ **ii)** $Mn^{2+} + S_2O_8^{2-}$

iii) $Mn^{2+} + NaBiO_3 + HNO_3$ **iv)** $Mn^{2+} + PbO_2 + HNO_3$

v) $K_2MnO_4 + Cl_2$ **vi)** $ZnCl_2 + H_2S + NH_4OH(aq.)$

38. Consider the following reaction sequence.

$$[PtCl_4]^{2-} \xrightarrow[-2Cl^-]{+2Py} P \xrightarrow[-Cl^-]{+NH_3} Q \xrightarrow[-Py]{+NH_3} R$$

Find the sum of number of stereoisomers of complexes P, Q and R.

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MATHEMATICS

Max. Marks: 60

SECTION-1(Maximum Marks: 24)

One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks : 0 If unanswered;
Negative Marks : −2 In all other cases.

39. Let $f:R^+ \to R, f(\lambda)=1$, where $\lambda \in R^+ (\lambda \neq 1)$ and

$$f(x)f(y)+f\left(\frac{\lambda}{x}\right)f\left(\frac{\lambda}{y}\right)=2f(xy)\forall x,y\in R^+ \text{ then}$$

A) Range of $f(x)$ contains only two integers

B) $f(x)$ is periodic function

C) $\sum_{r=-50}^{50} f\left(\frac{51-r}{101-r}\right)=101, r\in I$

D) $f(x)$ is bounded function

40. For non negative integers s and r, let

$$\binom{s}{r}=\begin{cases}\dfrac{s!}{r!(s-r)!} & \text{if } r\le s,\\ 0 & \text{if } r>s.\end{cases}$$

For positive integers m and n, let

$$g(m,n)=\sum_{p=0}^{m+n}\frac{f(m,n,p)}{\binom{n+p}{p}}$$

Where for any nonnegative integer p.

$$f(m,n,p)=\sum_{i=0}^{p}\binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i}.$$

Then which of the following statements is/are TRUE?

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A) $g(m,n)=g(n,m)$ for all positive integers m, n

B) $g(m,n+1)=g(m+1,n)$ for all positive integers m, n

C) g(2m, 2n) = 2g(m, n)for all positive integers m, n

D) g(2m, 2n) = $(g(m,n))^2$ for all positive integers m, n

41. Let f be a function having a continuous derivative on [0,1] and with the property that $0<f'(x)\le 1$. Also suppose that $f(0)=0$.

Then choose the CORRECT options:

A) $\left(\int_0^1 f(x)dx\right)^2 \ge \int_0^1 (f(x))^3 dx$ **B)** $(f(x))^2 \ge 2\int_0^x f(x)dx$

C) $\int_0^1 f(x)dx \le \frac{1}{2}$ **D)** $\int_0^1 f(x)\cos x dx \le \frac{55}{144}$

42. The value of $\sum_{k=1}^{\infty}\frac{6^k}{(3^k-2^k)(3^{k+1}-2^{k+1})}$ can be equal to :

A) $\lim_{x\to 0}\frac{1-\cos 2x}{x^2}$

B) $\lim_{n\to\infty}\left(\sin\frac{\pi}{2n}\times\sin\frac{2\pi}{2n}\times\sin\frac{3\pi}{2n}.....\sin\frac{(n-1)\pi}{n}\right)^{1/n}$

C) $\frac{\int_{\pi/2}^{0}\ln|\cos 2x|dx}{\pi\ln 2}$

D) $\frac{2}{\pi\ln 2}\int_0^{\pi/4}\ln(1+\tan x)dx$

43. Let a be a positive integer such that the limit $\lim_{x\to 1}\left(\frac{1}{x-1}-\frac{1}{x^a-2x+1}\right)$ exists and is equal to b : (where $b\ne 0$), then which of the following is/are correct

A) $\tan^{-1}(\tan a)$ is equal to $3-\pi$

B) $\tan^{-1}(\tan b)$ is equal to $3-\pi$

C) $\tan^{-1}(\tan(a+b))$ is equal to $5-2\pi$

D) $\tan^{-1}(\tan(a-b))$ is equal to 1.

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44. Let $I_1=\int_0^1 \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)dx; I_2=\int_0^1 \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)dx$ and

$I_3=\lim_{x\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+.....\frac{1}{2n}\right)$, then :

A) $I_1+I_3=I_2$ **B)** $I_1+I_2+I_3=\ln 2$

C) $I_2+I_3=I_1$ **D)** $I_1+I_2+I_3=\frac{\pi}{2}+\ln 2$

SECTION-2(Maximum Marks: 12)
Paragraph with Numerical

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 45 and 46

Question Stem

$f(x)$ is a differentiable function and

If $x\,f(x)=3f^2(x)+2$, for $x>2\sqrt{6}; f(x)>\sqrt{2/3}$ and

$I(x)=\int\frac{2x^2-12xf(x)+f(x)}{\left(6f(x)-x\right)\left(x^2-f(x)\right)^2}dx+c$

45. If $I(7)=\frac{1}{47}$, then value of C is :

46. If $\lim_{x\to\infty}6\frac{f(x)}{x}$ exists, then sum of its possible value(s) is

Question Stem for Question Nos. 47 and 48

Question Stem

Let a function is defined by $f:R-\{0\}\to R-\{0\}$ A line $\frac{x}{-2}=\frac{-f(t)y}{t}=t^2z$ is perpendicular to the line of intersection of the planes.

$P_1: tf(t)x+f\left(\frac{1}{t^2}\right)z+f(-t)=0$

$P_2: ty+f(-t)z+f(t^2)=0$

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47. Let $t=\tan\theta, \theta \in R-\left\{(2n+1)\frac{\pi}{2}, n\pi, n\pi\pm\frac{\pi}{4}\right\}, n\in I$ such that $f(\tan\theta)+\tan(\lambda\theta).f(\cot^2\theta)=0$, then value of λ is equal to : (where $\lambda \in N$)

48. Let $f(t)$ is an identify function in the domain. If a plane is obtained by rotating P_2 about its line of intersection with P by 90 and this new plane passes through origin then sum of values of 't' is equal to :

Question Stem for Question Nos. 49 and 50

Question Stem

Consider, $f(x)=\left|\sin\left((2r_1-1)\frac{\pi}{6}\right)x\right|+\left|\cos\left(\frac{r_2\pi}{6}\right)x\right|$.

Let a fair dice is thrown twice and r_1, r_2 are the numbers obtained on the dice in its first and second throw respectively.

49. The probability that $f(1)$ is an integer, is:

50. The probability that f (2) is irrational, is:

SECTION-3(Maximum Marks: 12)
Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-I:

Let $f(x)=x^{2010}+x^{1010}-x^{510}+x^{210}+x^2$. If $f(x)$ is divided by $x^2(x^2-1)$, then we get remainder as $g(x)$, function of x.

51. If $g(x)$ is defined from R to R, then:

A) g(x) is injective but not surjective

B) g(x) is surjective but not injective

C) g(x) is neither injective nor surjective

D) g(x) is injective and surjective

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52. If roots of $g(x)=0$ lies between the roots of the equation $x^2-2(a+1)x+a(a-1)=0$ then number of integral values of a will be

A) 0 **B)** 1 **C)** 2 **D)** 3

Paragraph-II:

Let $C_1: x^2+y^2=r^2$ and $C_2:(x-p)^2+(y-q)^2=r^2$ be 2 circle with radius $r(r>0)$ and have n points of intersection, (x_i,y_i) for $i\in\{1,2,.....,n\}$. If C_1 and C_2 are orthogonal at all points of intersection, then:

53. As we move the centre of C_2 along $p+bq=0$ for some constant $b\neq 0$, then $\frac{dr}{dq}$ is equal to

A) $\frac{b^2q+q}{r}$ **B)** $\frac{b^2q+q}{2r}$ **C)** $\frac{b^2q+q}{3r}$ **D)** $\frac{b^2q+q}{4r}$

54. As we move the centre of C_2 along the curve $q=a$ for some constant $a\neq 0$, then $\frac{dr}{dp}$ is equal to :

A) $\frac{p}{4r}$ **B)** $\frac{p}{3r}$ **C)** $\frac{p}{2r}$ **D)** $\frac{p}{r}$

SECTION-4(Maximum Marks: 12)
Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct integer is entered;
  Zero Marks : 0 In all other cases.

55. For every positive real number x, $g(x)=\lim_{r\to 0}\left((x+1)^{r+1}-x^{r+1}\right)^{1/r}$. If $L=\lim_{x\to\infty}\frac{g(x)}{x}$; then value of $[L]$ is. Where $[.]$ is greatest integer function.

56. Let $n\in N$, $A^n=\begin{bmatrix}2 & 1\\ 1 & 0\end{bmatrix}^n=\begin{bmatrix}a_{11}(n) & a_{12}(n)\\ a_{21}(n) & a_{22}(n)\end{bmatrix}$. If $\lim_{n\to\infty}\frac{a_{12}(n)}{a_{22}(n)}=L$, where $L^2=\sqrt{a}+\sqrt{b},(a,b\in N,a>b)$ then $(a+b)$ is equal to

57. Consider $f(x)=x^4+ax^3+bx^2+cx+d$. If straight line $y=3x+2$ is tangent to the curve $y=f(x)$ at $P(1,f(1))$ which again intersects $f(x)$ at $Q(2,8)$ and $f''(1)=0$ then find the value of $f(3)$

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BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

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ALL INDIA OPEN CATEGORY RANKS



BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

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